

Chapter 3 : In Class Exercise

1. Determine whether goal-driven or data-driven search would be preferable for solving each of the following problems. Justify your answer.

- a) Diagnosing **mechanical problems** in an automobile. *goal-driven*
 "the car got problem"
 we search the causes by backward searching
- b) You have met a person who claims to be your distant cousin, **with a common ancestor named John Doe**. You would like to verify her claim. *goal-driven*
 already know the goal "Is he really common ancestor?"
 now need to check backward
- c) Another person claims to be your distant cousin. He does not know the common ancestor's name but **knows that it was no more than eight generations back**. You would like to either find this ancestor or determine that she did not exist. *data-driven*
 data given "eight-generation back"
- d) A theorem **prover for plane geometry**. *goal-driven*
 the goal "is it prover for plane geometry" prove it
- e) A program for examining sonar readings and interpreting them, such as telling a large submarine from a small submarine from a whale from a school of fish. *data-driven*
 data given
- f) An expert system that will help a human classify plants by species, genus, etc. *data-driven*
 data

2. Apply the backtrack algorithm on the graph in **Figure 1**. Begin from state A. Keep track of the successive values of NSL, SL, CS, etc.

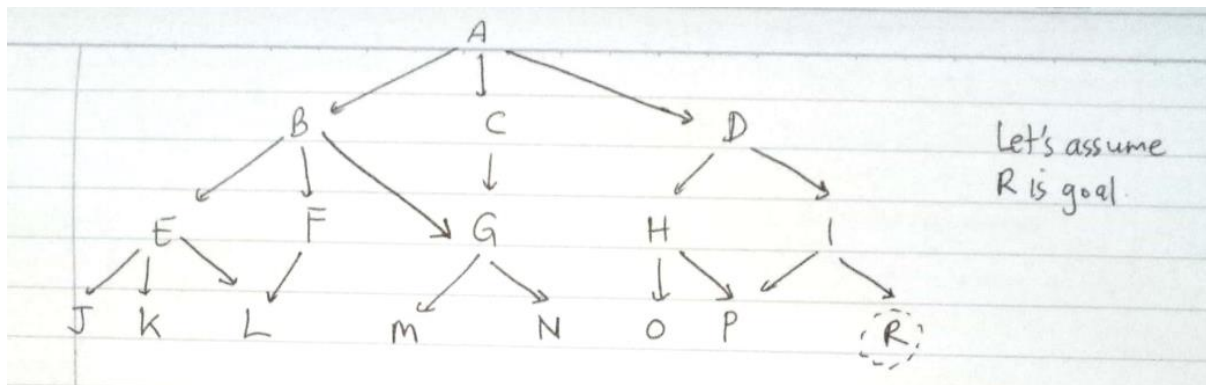


Figure 1

After Iteration	CS (Current State)	SL (State List)	NSL (New State List)	DE (Dead End)
0	A	[A]	[A]	[]
1	B	[B A]	[B C D A]	[]
2	E	[E B A]	[E F G B C D A]	[]
3	J	[J E B A]	[J K L E F G B C D A]	[]
4	K	[K E B A]	[K L E F G B C D A]	[J]
5	L	[L E B A]	[L E F G B C D A]	[K J]
6	F	[F B A]	[F G B C D A]	[E L K J]
7	G	[G B A]	[G B C D A]	[F E L K J]
8	M	[M G B A]	[M N A B C D A]	[F E L K J]
9	N	[N G B A]	[N G B C D A]	[M F E L K J]
10	C	[C A]	[C D A]	[B G N M F E L K J]
11	D	[D A]	[D A]	[C B G N M F E L K J]
12	H	[H D A]	[H I D A]	[C B G N M F E L K J]
13	O	[O H D A]	[O P H I D A]	[C B G N M F E L K J]
14	P	[P H O A]	[P H I D A]	[O C B G N M F E L K J]
15	I	[I D A]	[I D A]	[H P O C B G N M F E L K J]
16	R	[R I D A]	[R I D A]	

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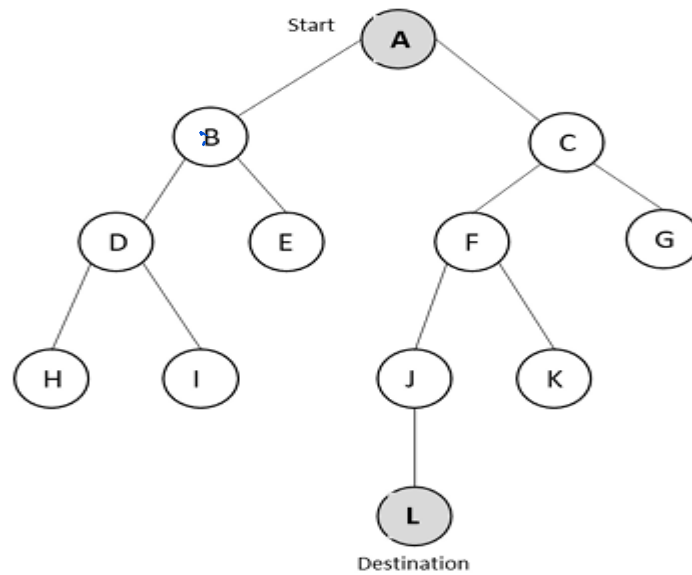


Figure 3.0

7. You are applying **Depth-First Search (DFS)** to find a path from **node A to node L** (see Figure 3.0). Initialize the open list, closed list, and start the traversal from node A

- Trace the search tree until you reach the destination (node L). Indicate the open list, closed list, and the current node at each step/iteration.

	Current node
1. open = [A] ; closed = []	A
2. open = [B, C] ; closed = [A]	B
3. open = [D, E, C] ; closed = [B, A]	D
4. open = [H, I, E, C] ; closed = [D, B, A]	H
5. open = [I, E, C] ; closed = [H, D, B, A]	I
6. open = [E, C] ; closed = [I, H, D, B, A]	E
7. open = [C] ; closed = [E, I, H, D, B, A]	C
8. open = [F, G] ; closed = [C, E, I, H, D, B, A]	F
9. open = [J, K, G] ; closed = [F, C, E, I, H, D, B, A]	J
10. open = [L, K, G] ; closed = [J, F, C, E, I, H, D, B, A]	L

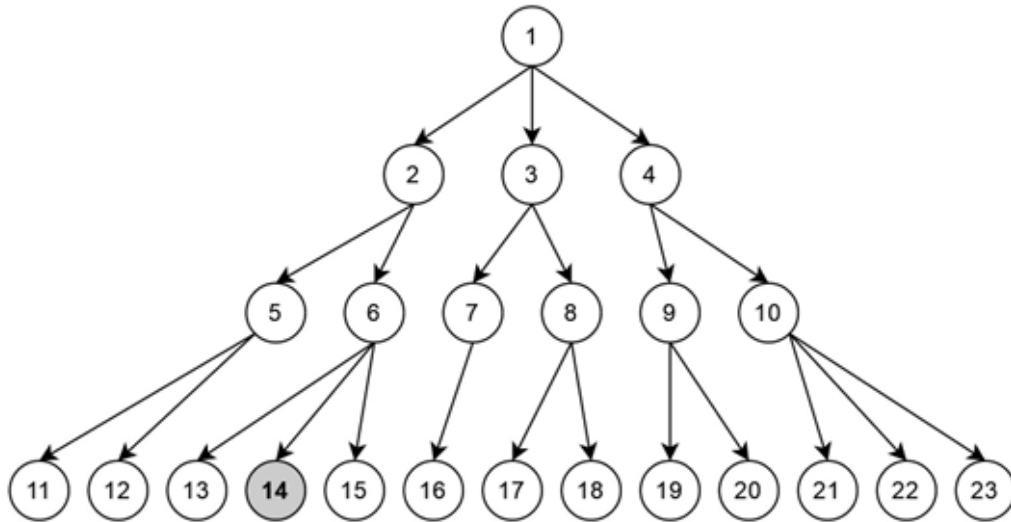


Figure 2.0

6. a. Perform Breadth-First Search on the state space graph in Figure 2.0 above. List down the order of nodes visited from starting node (Node 1) to the goal (Node 14) in a table as shown in Table 1.0.

* You may add more row if necessary.

(13 marks)

Table 1.0

Iteration	Open List	Closed List
0	1	[]
1	2, 3, 4	[1]
2	3, 4, 5, 6	[1, 2]
3	4, 5, 6, 7, 8	[1, 2, 3]
4	5, 6, 7, 8, 9, 10	[1, 2, 3, 4]
5	6, 7, 8, 9, 10, 11, 12	[1, 2, 3, 4, 5]
6	7, 8, 9, 10, 11, 12, 13, 14	[1, 2, 3, 4, 5, 6]